Technical Note: Design Theory for the Butterworth-Optimized Photodiode Transimpedance Amplifier

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This document provides a self-contained description of the mathematical theory implemented in the photodiode transimpedance amplifier (TIA) design tool. It covers the single-pole opamp model, the derivation of the closed-loop second-order transfer function, the exact Butterworth-optimum feedback capacitance without the common $\omega_c R_f C_f \gg 1$ approximation, the $-3\,\mathrm{dB}$ bandwidth, the noise equivalent bandwidth for a second-order Butterworth response, the piecewise noise-gain integration for opamp voltage noise, and the shot/thermal noise analysis. All algebraic steps are shown explicitly.

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1. Circuit model and assumptions

Circuit topology

The TIA is an inverting opamp configuration with a photodiode at the input. The non-inverting input is connected to ground.

- I_{pd}: photocurrent from the photodiode (signal source, modelled as ideal current source)
- R_f, C_f: feedback resistance and capacitance (in parallel), connected from the output to the inverting input
- C_i: total input capacitance at the inverting node, $C_i = C_d + C_{icm} + C_{id}$
  - C_d: photodiode junction capacitance
  - C_{icm}: opamp common-mode input capacitance
  - C_{id}: opamp differential input capacitance
- C_s: stray (parasitic) capacitance at the inverting node to ground

Opamp model

The opamp is modelled with a single-pole open-loop gain:

$$A(s) = \frac{A_0\,\omega_p}{s + \omega_p} \qquad (1.1)$$

where A_0 is the DC open-loop gain and ω_p is the dominant pole frequency. For frequencies well above the dominant pole ($s \gg \omega_p$, i.e. $f \gg f_p$, typically satisfied for $f > 10$ Hz), this simplifies to:

$$A(s) \approx \frac{A_0\,\omega_p}{s} = \frac{\omega_c}{s} \qquad (1.2)$$

where $\omega_c = A_0\,\omega_p = 2\pi f_c$ is the gain-bandwidth product (GBW) in rad/s. This "integrator model" is the standard approximation used throughout TIA design literature and is accurate across the entire frequency range of interest (kHz to MHz).

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2. Closed-loop transimpedance transfer function

Goal

Derive the closed-loop transimpedance:

$$Z_{\mathrm{TI}}(s) \equiv \frac{V_{\mathrm{out}}(s)}{I_{\mathrm{pd}}(s)}.$$

Step 1: Define the feedback impedance

R_f and C_f are in parallel:

$$Z_f = R_f \parallel \frac{1}{sC_f} = \frac{R_f}{1 + sR_fC_f} \qquad (2.1)$$

Step 2: Opamp constraint

With the non-inverting input grounded:

$$V_{\mathrm{out}} = -A(s) \cdot V^- = -\frac{\omega_c}{s}\,V^-$$

Solving for V^-:

$$V^- = -\frac{s}{\omega_c}\,V_{\mathrm{out}} \qquad (2.2)$$

Step 3: Kirchhoff's current law at the inverting node

All currents leaving the inverting node must sum to zero. The current into the node from the photodiode is I_{pd}. The currents leaving the node are:

- Through the feedback network to the output: $(V^- - V_{\mathrm{out}})/Z_f$
- Through the total shunt capacitance $(C_i + C_s)$ to ground: $V^- \cdot s(C_i + C_s)$

KCL:

$$I_{\mathrm{pd}} + \frac{V^- - V_{\mathrm{out}}}{Z_f} + V^- \cdot s(C_i + C_s) = 0$$

Note the sign convention: current flowing from the node through Z_f toward V_{out} is $(V^- - V_{\mathrm{out}})/Z_f$. Rearranging with current into the node on the left:

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}} - V^-}{Z_f} - V^- \cdot s(C_i + C_s) \qquad (2.3)$$

Step 4: Substitute the opamp constraint

Substitute (2.1) and (2.2) into (2.3).

First, compute $V_{\text{out}} - V^-$:

$$V_{\text{out}} - V^- = V_{\text{out}} - \left(-\frac{s}{\omega_{\text{c}}} V_{\text{out}}\right) = V_{\text{out}} \left(1 + \frac{s}{\omega_{\text{c}}}\right) = V_{\text{out}} \cdot \frac{\omega_{\text{c}} + s}{\omega_{\text{c}}} \tag{2.4}$$

Substitute into the feedback current term:

$$\frac{V_{\text{out}} - V^-}{Z_{\text{f}}} = V_{\text{out}} \cdot \frac{\omega_{\text{c}} + s}{\omega_{\text{c}}} \cdot \frac{1 + sR_{\text{f}}C_{\text{f}}}{R_{\text{f}}} \tag{2.5}$$

The shunt capacitance current term:

$$-V^- \cdot s(C_{\text{i}} + C_{\text{s}}) = \frac{s}{\omega_{\text{c}}} V_{\text{out}} \cdot s(C_{\text{i}} + C_{\text{s}}) = \frac{s^2(C_{\text{i}} + C_{\text{s}})}{\omega_{\text{c}}} V_{\text{out}} \tag{2.6}$$

Step 5: Combine

Substituting (2.5) and (2.6) into (2.3):

$$I_{\text{pd}} = V_{\text{out}} \left[\frac{(\omega_{\text{c}} + s)(1 + sR_{\text{f}}C_{\text{f}})}{\omega_{\text{c}} R_{\text{f}}} + \frac{s^2(C_{\text{i}} + C_{\text{s}})}{\omega_{\text{c}}}\right]$$

Factor out $1/\omega_{\text{c}}$:

$$I_{\text{pd}} = \frac{V_{\text{out}}}{\omega_{\text{c}}} \left[\frac{(\omega_{\text{c}} + s)(1 + sR_{\text{f}}C_{\text{f}})}{R_{\text{f}}} + s^2(C_{\text{i}} + C_{\text{s}})\right] \tag{2.7}$$

Step 6: Expand the product

$$(\omega_{\text{c}} + s)(1 + sR_{\text{f}}C_{\text{f}}) = \omega_{\text{c}} + \omega_{\text{c}} sR_{\text{f}}C_{\text{f}} + s + s^2 R_{\text{f}}C_{\text{f}}$$

Rearranging in powers of s:

$$= s^2 R_{\text{f}}C_{\text{f}} + s(1 + \omega_{\text{c}} R_{\text{f}}C_{\text{f}}) + \omega_{\text{c}} \tag{2.8}$$

Step 7: Divide by R_{f} and add the shunt term

$$\frac{(\omega_{\text{c}} + s)(1 + sR_{\text{f}}C_{\text{f}})}{R_{\text{f}}} + s^2(C_{\text{i}} + C_{\text{s}})$$

$$= s^2 C_{\text{f}} + \frac{s(1 + \omega_{\text{c}} R_{\text{f}}C_{\text{f}})}{R_{\text{f}}} + \frac{\omega_{\text{c}}}{R_{\text{f}}} + s^2(C_{\text{i}} + C_{\text{s}})$$

$$= s^2(C_{\text{i}} + C_{\text{f}} + C_{\text{s}}) + s\,\frac{1 + \omega_{\text{c}} R_{\text{f}}C_{\text{f}}}{R_{\text{f}}} + \frac{\omega_{\text{c}}}{R_{\text{f}}} \tag{2.9}$$

Step 8: Form the transfer function

From (2.7) and (2.9):

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}}}{\omega_{\mathrm{c}}} \left[s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}} \right]$$

$$Z_{\mathrm{TI}}(s) = \frac{V_{\mathrm{out}}}{I_{\mathrm{pd}}} = \frac{\omega_{\mathrm{c}}}{s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\frac{1+\omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}}} \tag{2.10}$$

(The overall sign is negative for the inverting configuration; we write the magnitude here and note the 180 degree phase inversion separately.)

Step 9: Identify the standard second-order form

A standard second-order low-pass transfer function has the form:

$$H(s) = \frac{K\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2}$$

Dividing numerator and denominator of (2.10) by $(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})$:

$$Z_{\mathrm{TI}}(s) = \frac{\frac{\omega_{\mathrm{c}}}{C_{\mathrm{i}}+C_{\mathrm{f}}+C_{\mathrm{s}}}}{s^2 + \frac{1+\omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}}+C_{\mathrm{f}}+C_{\mathrm{s}})}\,s + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}}+C_{\mathrm{f}}+C_{\mathrm{s}})}} \tag{2.11}$$

By comparison with the standard form, we identify:

$$\omega_{\mathrm{n}}^2 = \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{2.12}$$

$$2\zeta\omega_{\mathrm{n}} = \frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{2.13}$$

The DC gain ($s = 0$) of (2.11):

$$Z_{\mathrm{TI}}(0) = \frac{\omega_{\mathrm{c}}/(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{\omega_{\mathrm{n}}^2} = \frac{\omega_{\mathrm{c}}/(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}/[R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})]} = R_{\mathrm{f}}.$$

The numerator constant can therefore be written as $R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2$, giving the final form (restoring the sign for the inverting configuration):

$$Z_{\mathrm{TI}}(s) = \frac{-R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2} \tag{2.14}$$

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3. Butterworth-optimum feedback capacitance

Objective

Find the value of C_{f} that makes the closed-loop response maximally flat (Butterworth).

Step 1: Butterworth condition

A second-order Butterworth (maximally flat magnitude) filter has quality factor:

$$Q = \frac{1}{\sqrt{2}}$$

The relationship between Q and the damping ratio ζ is $Q = 1/(2\zeta)$, so:

$$\zeta = \frac{1}{\sqrt{2}} \qquad (3.1)$$

This is the unique damping ratio that produces zero gain peaking in the magnitude response (i.e. $|Z_{\mathrm{TI}}(j\omega)|$ is monotonically decreasing).

Step 2: Translate the condition to circuit parameters

The Butterworth condition (3.1) implies:

$$(2\zeta\omega_{\mathrm{n}})^2 = 4\zeta^2\omega_{\mathrm{n}}^2 = 4 \cdot \frac{1}{2} \cdot \omega_{\mathrm{n}}^2 = 2\,\omega_{\mathrm{n}}^2.$$

Substituting (2.12) and (2.13):

$$\left(\frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}\right)^2 = 2 \cdot \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}$$

Multiply both sides by $[R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})]^2$:

$$(1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}})^2 = 2\,\omega_{\mathrm{c}} R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) \qquad (3.2)$$

Step 3: Solve for C_{f}

Define a characteristic capacitance:

$$C_{\mathrm{c}} \equiv \frac{1}{\omega_{\mathrm{c}} R_{\mathrm{f}}} = \frac{1}{2\pi f_{\mathrm{c}} R_{\mathrm{f}}} \qquad (3.3)$$

This has a physical interpretation: C_{c} is the capacitance whose impedance at f_{c} equals R_{f}.

Rewrite (3.2) using $\omega_{\mathrm{c}} R_{\mathrm{f}} = 1/C_{\mathrm{c}}$:

$$\left(1 + \frac{C_{\mathrm{f}}}{C_{\mathrm{c}}}\right)^2 = \frac{2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{C_{\mathrm{c}}}$$

Expand the left side:

$$1 + \frac{2C_{\mathrm{f}}}{C_{\mathrm{c}}} + \frac{C_{\mathrm{f}}^2}{C_{\mathrm{c}}^2} = \frac{2(C_{\mathrm{i}} + C_{\mathrm{s}})}{C_{\mathrm{c}}} + \frac{2C_{\mathrm{f}}}{C_{\mathrm{c}}}$$

The $2C_{\mathrm{f}}/C_{\mathrm{c}}$ terms cancel:

$$1 + \frac{C_{\mathrm{f}}^2}{C_{\mathrm{c}}^2} = \frac{2(C_{\mathrm{i}} + C_{\mathrm{s}})}{C_{\mathrm{c}}}$$

Multiply through by C_c^2:

$$C_c^2 + C_f^2 = 2(C_i + C_s)C_c$$

$$C_f^2 = 2(C_i + C_s)C_c - C_c^2 = C_c\big(2(C_i + C_s) - C_c\big) \qquad (3.4)$$

Taking the positive root:

$$C_f = \sqrt{C_c\big(2(C_i + C_s) - C_c\big)}, \qquad C_c = \frac{1}{2\pi f_c R_f} \qquad (3.5)$$

Validity condition

Equation (3.5) requires the argument of the square root to be non-negative:

$$2(C_i + C_s) - C_c > 0 \quad \Longleftrightarrow \quad C_i + C_s > \frac{C_c}{2} = \frac{1}{2\omega_c R_f}.$$

When this condition is violated, the opamp GBW is too low (or R_f too small) to achieve a Butterworth response — the circuit is fundamentally underdamped regardless of C_f.

Relation to the common approximation

Many references simplify by assuming $\omega_c R_f C_f \gg 1$ (equivalently $C_f \gg C_c$). Under this approximation, (3.2) becomes:

$$(\omega_c R_f C_f)^2 \approx 2\,\omega_c R_f\,(C_i + C_s) \quad \Longrightarrow \quad C_f \approx \sqrt{\frac{2(C_i + C_s)}{\omega_c R_f}} = \sqrt{2(C_i + C_s)C_c}.$$

Equation (3.5) is the **exact** solution that does not rely on this approximation. The difference is significant when C_c is not negligible compared to $C_i + C_s$ (i.e. when R_f is small or f_c is low).

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4. Bandwidth

The $-3\,\mathrm{dB}$ bandwidth of a second-order Butterworth

For a general second-order transfer function with damping ratio ζ, the squared magnitude response is:

$$|H(j\omega)|^2 = \frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}$$

For the Butterworth case $\zeta = 1/\sqrt{2}$, substituting and simplifying:

$$|H(j\omega)|^2 = \frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + 2\omega_n^2\omega^2}$$

Expand the denominator:

$$= \omega_n^4 - 2\omega_n^2\omega^2 + \omega^4 + 2\omega_n^2\omega^2 = \omega_n^4 + \omega^4$$

Therefore:

$$|H(j\omega)|^2 = \frac{\omega_\mathrm{n}^4}{\omega_\mathrm{n}^4 + \omega^4} = \frac{1}{1 + (\omega/\omega_\mathrm{n})^4} \qquad (4.1)$$

This is the defining form of a second-order Butterworth response.

At $\omega = \omega_\mathrm{n}$:

$$|H(j\omega_\mathrm{n})|^2 = \frac{1}{1+1} = \frac{1}{2} \quad \Longrightarrow \quad |H(j\omega_\mathrm{n})| = \frac{1}{\sqrt{2}}\ (-3\,\mathrm{dB}).$$

Therefore the $-3\,\mathrm{dB}$ bandwidth is exactly the natural frequency:

$$\omega_{-3\,\mathrm{dB}} = \omega_\mathrm{n}.$$

In terms of ordinary frequency:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_\mathrm{n}}{2\pi} \qquad (4.2)$$

Express in terms of circuit parameters

From (2.12):

$$\omega_\mathrm{n} = \sqrt{\frac{\omega_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}} = \sqrt{\frac{2\pi f_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}}$$

Therefore:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_\mathrm{n}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{2\pi f_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}} = \underline{\sqrt{\frac{f_\mathrm{c}}{2\pi R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}}} \qquad (4.3)$$

5. Noise equivalent bandwidth

Definition

The noise equivalent bandwidth (NEB) of a filter $H(f)$ is the bandwidth of an ideal rectangular (brick-wall) filter with the same peak gain that passes the same total noise power:

$$\mathrm{NEB} = \frac{1}{|H_\mathrm{max}|^2} \int_0^\infty |H(f)|^2\, df$$

For a Butterworth filter, $|H_\mathrm{max}| = |H(0)| = 1$, so:

$$\mathrm{NEB} = \int_0^\infty |H(f)|^2\, df \qquad (5.1)$$

General n-th order Butterworth NEB

For an n-th order Butterworth filter, the squared magnitude response is:

$$|H(f)|^2 = \frac{1}{1 + (f/f_{-3\,\mathrm{dB}})^{2n}}$$

Substituting into (5.1) with $u = f/f_{-3\,\mathrm{dB}}$:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \int_0^\infty \frac{du}{1 + u^{2n}} \qquad (5.2)$$

The integral $I_n = \displaystyle\int_0^\infty \frac{du}{1 + u^{2n}}$ is a standard result that can be evaluated using the substitution $t = u^{2n}$ and recognizing the beta function.

Setting $t = u^{2n}$, so $u = t^{1/(2n)}$, $du = \frac{1}{2n}\, t^{1/(2n)-1}\, dt$:

$$I_n = \int_0^\infty \frac{1}{1+t} \cdot \frac{1}{2n}\, t^{1/(2n)-1}\, dt = \frac{1}{2n}\, B\!\left(\frac{1}{2n},\ 1 - \frac{1}{2n}\right)$$

where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is the beta function.

Using the reflection formula $\Gamma(z)\,\Gamma(1 - z) = \pi/\sin(\pi z)$ with $z = 1/(2n)$:

$$I_n = \frac{1}{2n} \cdot \frac{\pi}{\sin\!\left(\frac{\pi}{2n}\right)} \qquad (5.3)$$

Therefore:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2n\,\sin\!\left(\frac{\pi}{2n}\right)} \qquad (5.4)$$

Specialization to $n = 2$

For the second-order Butterworth ($n = 2$):

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2 \cdot 2 \cdot \sin\!\left(\frac{\pi}{4}\right)} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{4 \cdot \frac{\sqrt{2}}{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sqrt{2}}$$

Rationalizing:

$$= f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4}$$

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4} \approx 1.111\, f_{-3\,\mathrm{dB}} \qquad (5.5)$$

Comparison with the single-pole approximation

For a single-pole (first-order) system ($n = 1$):

$$\mathrm{NEB}_{n=1} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sin(\pi/2)} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2} \approx 1.571\, f_{-3\,\mathrm{dB}}.$$

Many TIA references use this single-pole approximation even when the circuit has a second-order response. The second-order Butterworth NEB is $1.111/1.571 = 0.707$ times the single-pole value — a 29% difference that directly affects computed noise levels.

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6. Noise analysis

The TIA output noise is the root-sum-square (RSS) of several independent noise sources. Each source has a spectral density that is shaped by the closed-loop gain before appearing at the output. The noise sources fall into two categories: (a) voltage noise of the opamp, which is amplified by the frequency-dependent noise gain, and (b) current/thermal noise sources whose spectra are flat and are shaped only by the closed-loop transimpedance $|Z_{\mathrm{TI}}(f)|^2$.

6.1 Noise gain

The noise gain $G_{\mathrm{n}}(f)$ is the closed-loop gain from the opamp's input voltage noise source e_{ni} to the output. For a TIA with feedback impedance $Z_{\mathrm{f}} = R_{\mathrm{f}} \| (1/sC_{\mathrm{f}})$ and total input-to-ground capacitance $(C_{\mathrm{i}} + C_{\mathrm{s}})$, the noise gain is:

$$G_{\mathrm{n}}(f) = 1 + \frac{Z_{\mathrm{in}}}{Z_{\mathrm{f}}} = 1 + \frac{C_{\mathrm{i}} + C_{\mathrm{s}}}{C_{\mathrm{f}} + C_{\mathrm{s}}} \cdot \frac{1 + sR_{\mathrm{f}}(C_{\mathrm{f}} + C_{\mathrm{s}})}{1 + sR_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}$$

This gain exhibits four distinct frequency regions that are exploited for piecewise integration:

1. **Low-frequency flat region** ($f < f_{\mathrm{zf}}$): $G_{\mathrm{n}} \approx 1$
2. **Rising region** ($f_{\mathrm{zf}} < f < f_{\mathrm{pf}}$): $G_{\mathrm{n}} \approx f/f_{\mathrm{zf}}$
3. **Plateau region** ($f_{\mathrm{pf}} < f < f_{\mathrm{i}}$): $G_{\mathrm{n}} \approx 1 + C_{\mathrm{i}}/(C_{\mathrm{f}} + C_{\mathrm{s}})$
4. **Roll-off region** ($f > f_{\mathrm{i}}$): G_{n} decreases as the loop gain falls below unity

where the characteristic frequencies are:

$$f_{\mathrm{zf}} = \frac{1}{2\pi R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \qquad (6.1)$$

$$f_{\mathrm{pf}} = \frac{1}{2\pi R_{\mathrm{f}}(C_{\mathrm{f}} + C_{\mathrm{s}})} \qquad (6.2)$$

$$f_{\mathrm{i}} = f_{\mathrm{c}} \cdot \frac{C_{\mathrm{f}} + C_{\mathrm{s}}}{C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}} \qquad (6.3)$$

Here f_{zf} is the noise-gain zero frequency, f_{pf} is the noise-gain pole frequency, and f_{i} is the unity loop-gain (intersection) frequency.

6.2 Opamp voltage noise (piecewise integration)

The opamp input voltage noise spectral density has a $1/f$ component below the corner frequency f_f and a white (flat) floor e_nif above it:

$$e_\mathrm{ni}(f) = e_\mathrm{nif}\sqrt{1 + f_\mathrm{f}/f} \qquad (6.4)$$

The output-referred voltage noise RMS is obtained by integrating the product of $e_\mathrm{ni}^2(f)$ and $G_\mathrm{n}^2(f)$ over frequency, divided into five segments. Defining a low-frequency cutoff f_1 (set to 0.01 Hz in the implementation):

Region 1 — $1/f$ noise (f_1 to f_f), noise gain ≈ 1:

$$E_\mathrm{noe,1}^2 = e_\mathrm{nif}^2 \cdot f_\mathrm{f} \ln\left(\frac{f_\mathrm{f}}{f_1}\right) \qquad (6.5)$$

Region 2 — white noise, flat gain (f_f to f_zf), noise gain ≈ 1:

$$E_\mathrm{noe,2}^2 = e_\mathrm{nif}^2 (f_\mathrm{zf} - f_\mathrm{f}) \qquad (6.6)$$

Region 3 — white noise, rising gain (f_zf to f_pf), noise gain $\approx f/f_\mathrm{zf}$:

$$E_\mathrm{noe,3}^2 = \frac{e_\mathrm{nif}^2}{f_\mathrm{zf}^2} \cdot \frac{f_\mathrm{pf}^3 - f_\mathrm{zf}^3}{3} \qquad (6.7)$$

Region 4 — white noise, plateau (f_pf to f_i), noise gain $\approx 1 + C_\mathrm{i}/(C_\mathrm{f} + C_\mathrm{s})$:

$$E_\mathrm{noe,4}^2 = e_\mathrm{nif}^2 \left(1 + \frac{C_\mathrm{i}}{C_\mathrm{f} + C_\mathrm{s}}\right)^2 (f_\mathrm{i} - f_\mathrm{pf}) \qquad (6.8)$$

Region 5 — above unity loop gain ($f > f_\mathrm{i}$), gain $\approx f_\mathrm{c}/f$:

$$E_\mathrm{noe,5}^2 = e_\mathrm{nif}^2 \cdot \frac{f_\mathrm{c}^2}{f_\mathrm{i}} \qquad (6.9)$$

Note: Eq. (6.9) arises from $\int_{f_\mathrm{i}}^{\infty} (f_\mathrm{c}/f)^2 \, df = f_\mathrm{c}^2/f_\mathrm{i}$.

The total output-referred opamp voltage noise is:

$$E_\mathrm{noe} = \sqrt{E_\mathrm{noe,1}^2 + E_\mathrm{noe,2}^2 + E_\mathrm{noe,3}^2 + E_\mathrm{noe,4}^2 + E_\mathrm{noe,5}^2} \qquad (6.10)$$

6.3 Feedback resistor thermal noise

The Johnson noise of R_f produces a current noise density $\sqrt{4k_\mathrm{B}T/R_\mathrm{f}}$, which flows through R_f to produce output voltage noise. Since this noise source sees the same transimpedance transfer function as the signal, the NEB (Eq. 5.5) is the appropriate bandwidth:

$$E_\mathrm{noR} = \sqrt{4k_\mathrm{B}TR_\mathrm{f} \cdot \mathrm{NEB}} \qquad (6.11)$$

6.4 Shot noise

Shot noise from DC currents (bias current I_b, dark current I_d, and signal photocurrent I_p) produces white current noise with spectral density $\sqrt{2qI}$. Each is converted to output voltage noise by the transimpedance R_f over the NEB:

$$E_\mathrm{noi,bias} = R_\mathrm{f}\sqrt{2q\,I_\mathrm{b}\cdot\mathrm{NEB}} \qquad (6.12)$$

$$E_\mathrm{noi,dark} = R_\mathrm{f}\sqrt{2q\,I_\mathrm{d}\cdot\mathrm{NEB}} \qquad (6.13)$$

$$E_\mathrm{noi,sig} = R_\mathrm{f}\sqrt{2q\,I_\mathrm{p}\cdot\mathrm{NEB}} \qquad (6.14)$$

6.5 Opamp current noise

The opamp input current noise floor i_nif (white spectral density) also flows through R_f:

$$E_\mathrm{noi,nif} = R_\mathrm{f}\,i_\mathrm{nif}\sqrt{\mathrm{NEB}} \qquad (6.15)$$

6.6 Total noise and signal-to-noise ratio

The total background noise (all sources except signal shot noise) is:

$$E_\mathrm{no,bg} = \sqrt{E_\mathrm{noe}^2 + E_\mathrm{noR}^2 + E_\mathrm{noi,bias}^2 + E_\mathrm{noi,dark}^2 + E_\mathrm{noi,nif}^2} \qquad (6.16)$$

The signal voltage is $V_\mathrm{sig} = R_\mathrm{f}\cdot I_\mathrm{p} = R_\mathrm{f}\cdot r_\phi\cdot P_\mathrm{opt}$, where r_ϕ is the photodiode responsivity and P_opt is the incident optical power. The signal-to-noise ratio is:

$$\mathrm{SNR} = \frac{V_\mathrm{sig}}{\sqrt{E_\mathrm{no,bg}^2 + E_\mathrm{noi,sig}^2}} \qquad (6.17)$$

The ratio $E_\mathrm{no,bg}/E_\mathrm{noi,sig}$ indicates whether the system is shot-noise limited (ratio < 1) or background-noise limited (ratio > 1).

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7. Implementation mapping

Theory	Implementation	File
Opamp model, Eq. (1.2)	`Opamp` dataclass (`f_c` field)	tia_design.py
Input capacitance C_i	`C_i = C_d + opamp.C_icm + opamp.C_id`	tia_design.py
Butterworth C_f, Eq. (3.5)	`C_f = (C_c * (2*(C_i+C_s) - C_c))**0.5`	tia_design.py
Bandwidth, Eq. (4.3)	`BW_t = (f_c / (2*pi*R_f*C_tot))**0.5`	tia_design.py
NEB, Eq. (5.5)	`NEB = (pi*sqrt(2)/4) * BW_t`	tia_design.py
Noise gain frequencies, Eqs. (6.1)–(6.3)	`f_zf`, `f_pf`, `f_i`	tia_design.py
Piecewise voltage noise, Eqs. (6.5)–(6.10)	`E_noe1` ... `E_noe5`, `E_noe`	tia_design.py
Thermal noise, Eq. (6.11)	`E_noR`	tia_design.py
Shot noise, Eqs. (6.12)–(6.14)	`E_noi_bias`, `E_noi_dark`, `E_noi_sig`	tia_design.py
Current noise, Eq. (6.15)	`E_noi_nif`	tia_design.py
SNR, Eq. (6.17)	`snr`	tia_design.py
Photodiode $C_\mathrm{d}(V_\mathrm{b})$ model	`Photodiode.C_d(Vb)`	tia_design.py
Multi-opamp comparison	`compare_opamps(...)`	tia_design.py

8. References

1. J. Graeme, *Photodiode Amplifiers: Op Amp Solutions*. McGraw-Hill, 1996. Ch. 5 (Eqs. 5.8–5.12).
2. P. C. D. Hobbs, *Building Electro-Optical Systems: Making It All Work*, 2nd ed. Wiley, 2009.

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